GERMÁN KÜBER

Senior Backend Engineer — Node.js, Python, .NET & Cloud-Native Systems

Las Palmas, Spain · EU Citizen — authorized to work across the EU/EEA, no visa sponsorship required
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PROFESSIONAL SUMMARY

Senior backend engineer and player-coach with 15+ years building production systems across **Node.js (TypeScript), Python (FastAPI, LangChain), C# / .NET, and Rust** — making architecture calls while staying deep in the code, and managing teams of up to 8 engineers. Recent hands-on work includes AI / LLM platforms with multi-provider orchestration, event-driven microservices, real-time pipelines, and serverless backends on AWS (Lambda, S3, SQS / SNS) and Azure. Comfortable across the full backend stack — REST and GraphQL APIs, async runtimes, distributed messaging (RabbitMQ, Apache Kafka, Azure Service Bus, AWS SNS / SQS), and relational and document databases (PostgreSQL, MySQL, MongoDB, Redis). Strong background in event-driven architecture, microservices, Domain-Driven Design, and clean architectural patterns. Microsoft MVP for 8 consecutive years (Developer Technologies) and Microsoft Certified: Azure Solutions Architect Expert (AZ-305). Works well in cross-functional remote teams, contributes to code review and architecture discussions, and ships with attention to detail and ownership. EU citizen (Spanish nationality) with full authorization to work across the EU/EEA — no visa or sponsorship required — open to self-funded relocation anywhere in the EU.

CORE EXPERTISE

Backend Languages & Runtimes: Node.js (TypeScript, Express, Fastify, NestJS), Python (FastAPI, async, LangChain), C# / .NET (.NET 6 / 7 / 8 / 9, ASP.NET Core, Entity Framework Core), Rust (tokio, async), SQL

AWS & Cloud Infrastructure: AWS Lambda, S3, SNS, SQS, EventBridge, ECS, RDS, Bedrock; Azure (App Service, Functions, AKS, Service Bus, Cosmos DB, API Management, Entra ID, Key Vault); GCP (Vertex AI); Docker, Kubernetes, Terraform, ARM templates, CI / CD pipelines

APIs & Integration: REST APIs, GraphQL, gRPC, OpenAPI / Swagger, OAuth 2.0 / OIDC, JWT, webhooks, API design at scale

Messaging & Distributed Systems: AWS SNS / SQS, Azure Service Bus, RabbitMQ, Apache Kafka, MassTransit, NServiceBus, event-driven architecture, async runtimes, eventual consistency, retry / dead-letter patterns, distributed transactions

Databases & Storage: PostgreSQL (pgvector), Neo4j, MySQL (Aurora), MongoDB, Cosmos DB, SQL Server / Azure SQL, Redis (cache, pub-sub, streams), DynamoDB, query optimization, schema design

Architecture Patterns: Microservices, event-driven architecture, Domain-Driven Design (DDD), CQRS, Event Sourcing, Saga patterns, Clean Architecture / Hexagonal, Vertical Slice, serverless patterns, polyglot persistence

Frontend (when needed): React, Next.js, Angular, TypeScript, reusable component libraries

AI / LLM Systems: LLM orchestration (OpenAI, Anthropic, Gemini, AWS Bedrock, Azure OpenAI, Vertex AI), MCP (Model Context Protocol) servers, LangChain, LangGraph, knowledge graphs (Neo4j), code graphs, RAG pipelines with vector databases (Pinecone, Chroma, pgvector, Redis), eval harnesses (LLM-as-judge, record-and-replay), real-time audio pipelines

Observability & DevOps: Prometheus, OpenTelemetry, Application Insights, structured logging, distributed tracing, on-call rotation, post-mortems, CI / CD (GitHub Actions, Azure DevOps), GitOps

Security & Identity: OAuth 2.0 / OIDC, Entra ID (Azure AD), JWT, API security (rate limiting, scopes), Zero Trust principles, OWASP Top 10

Programming Paradigms & Design: Object-Oriented Programming (OOP), Functional Programming, Design Patterns, Architecture Patterns, SOLID principles, data structures, algorithm design, clean code

Engineering Practices: Test-Driven Development (TDD), pair programming, code review, mentoring, technical interviewing, ADRs, design reviews, CI / CD, DevOps practices, Infrastructure as Code (Terraform), async remote collaboration

PROFESSIONAL EXPERIENCE

AI Architect

Nera — Feb 2026 – Present (full-time) — Remote

- Own Nera's AI architecture: the knowledge and code-intelligence layer that lets the whole company query the business in natural language, and the agentic tooling product and engineering use to go from problem to spec.
- Built a code graph of the entire microservice fleet (66 repos, 77k nodes, 210k relationships across Java, TypeScript and Go), enriched with runtime calls from OpenTelemetry and served through a 32-tool MCP server that engineers use to design features and product uses to write PRDs.
- Designed and built a company-wide ontology (28 classes, 26 invariants) with ingestion pipelines from Notion, Jira and GitHub into a Neo4j knowledge graph plus a pgvector semantic index, giving every area end-to-end traceability from problem to PR and natural-language Q&A over MCP.
- Built an AI product-engineering platform where multi-agent LangGraph flows take a problem through PAD, PRD, epic and tech spec, grounded in the code graph and business knowledge.
- Automated user-research capture with a pipeline that transcribes interviews and extracts verifiable quotes and insights into Notion with human approval, replacing a manual tracker.
- Made evals a release gate: record-and-replay harnesses, a 100-question benchmark for the knowledge graph, fidelity audits for the code graph and LLM-as-judge for generated PRDs.
- Drove Spec-Driven Development (SDD) across engineering with a shared framework (specs, TDD, review gates) that standardized how features are built and raised quality team-wide.

Independent AI Consultant — Agent Engineering & Enterprise AI Delivery

Self-employed (autónomo, Spain) — Mar 2026 – Jul 2026 — Remote, client under NDA

- Forward-deployed AI consultant embedded with an enterprise engineering organization: work directly with the customer to run discovery with business and technical stakeholders, architect the solution, and own delivery from pilot through production and continuous iteration.

- Design and ship production agentic workflows and backend services on LangChain, LangGraph and Microsoft Agent Framework, with eval frameworks and regression tests gating every change before production.
- Own the RAG platform end-to-end for internal company documentation — retrieval optimization, chunking, context engineering, and the data structuring work that makes internal knowledge consumable by LLMs.
- Cut inference cost and latency through semantic caching, semantic memory, and dynamic model routing across providers; deployed LoRA fine-tunes through Azure AI Foundry.

Senior Software Engineer & Co-Founder

KliverAI (Own Venture, closed Feb 2026) — Mar 2025 – Feb 2026 (full-time) — Remote, Spain

- Built and shipped backend services across Node.js (TypeScript) and Python (FastAPI, async) integrating three LLM providers (OpenAI, Anthropic, Google Gemini) with dynamic routing across direct APIs and managed cloud services (AWS Bedrock, Azure OpenAI, Vertex AI), picking the best model per task to cut inference spend.
- Rewrote the real-time layer from app-embedded SignalR to Azure SignalR Service with Redis-backed state, scaling concurrent users by 500% while holding sub-200ms latency on audio pipelines (OpenAI Realtime API).
- Developed event-driven serverless workflows on AWS Lambda with S3, SNS / SQS, and EventBridge for real-time agent pipelines and asynchronous LLM processing.
- Built production RAG pipelines and data stores across Pinecone, Chroma, pgvector, and Redis with semantic search, hybrid retrieval, and reranking strategies.
- Benchmarked TTS / STT services (ElevenLabs, Deepgram, AssemblyAI, OpenAI Whisper, Play.ht) on latency, cost, and quality — getting latency to 90ms and word accuracy to nearly 95%.
- Implemented MCP (Model Context Protocol) servers and clients to standardize tool integration across LLM workflows; built performance-critical components in Rust (tokio, async) for the audio and streaming layer.
- Instrumented LLM evaluation and observability with LangSmith, RAGAS, DeepEval, Langfuse, and OpenTelemetry GenAI semantic conventions; ran red-teaming and structured evals to reduce hallucination rates.
- Worked on a polyglot stack (Node.js, Python, .NET, Rust, Kubernetes, AWS); collaborated with a small distributed team of up to 3 engineers, ran code reviews and architectural discussions.

Fractional CTO

Rocking Product — Apr 2025 – Dec 2025 (part-time, parallel to KliverAI) — Remote

- Came in as Fractional CTO to an 8-engineer team and restructured it in the first quarter without stepping away from the code; ran technical interviews and hiring decisions.
- Built an image embeddings service (Azure AI Vision and CLIP) to improve product matching across brand comparisons, raising accuracy and lifting the client's revenue by 40%.
- Cut production bugs by 70% through team training and automated processes (testing, CI / CD, code review).
- Optimized cloud infrastructure and LLM usage through semantic caching, prompt compression, and dynamic model selection across providers, cutting combined cloud and API spend by 45%.
- Designed a reusable base architecture on LangChain and LangGraph with monitoring via Langfuse and LangSmith and prompt versioning; migrated the real-time LLM service from OpenAI to Azure AI Foundry for compliance and data retention, cutting latency and improving scalability.
- Introduced Spec-Driven Development (SDD), improving feature quality and delivery speed; proposed and led a full-stack migration from a fragmented JavaScript codebase to .NET.
- Built backend features in Node.js / TypeScript, Python (FastAPI, LangChain) and .NET (ASP.NET Core, Entity Framework Core); ran weekly public-speaking workshops to sharpen team communication, raising the quality of client meetings.

Lead Software Engineer (Distributed Systems)

Borderless Capital / CTF Capital — Sep 2021 – Mar 2025 — Remote

- Lead engineer for low-latency distributed systems in Rust (tokio, async) and Python / Cython, designing event-driven pipelines processing 10K+ opportunities a day from $50M+ in daily transactional flow and contributing $100K+ in attributable annual P&L with execution under 10ms.
- Migrated the system from Python / Cython to Rust and took it to 20x more transactions per second.
- Designed and built a production trading bot in .NET (C#), integrating with Rust execution components and Python signal layers; owned its full lifecycle including deployment and operations.
- Built transaction delivery infrastructure with parallel broadcasting across RPCs and direct validator / TPU routes, hitting sub-slot inclusion on Solana (~400ms); reverse-engineered Geth and Reth to improve transaction simulation and gas estimation, contributing to the REVM community.
- Owned 24 / 7 production reliability and incident response, including on-call rotation, post-mortems, and observability ownership with Prometheus, OpenTelemetry, and custom telemetry tracking 50+ real-time metrics.
- Ran technical due diligence on 5+ projects that ended up getting funded — architecture, risk, and feasibility — acting as the CTO's technical reference on the low-level side.
- Built Python / Cython FFI bindings between Rust performance-critical code and higher-level orchestration layers; contributed to code review, technical interviewing, and architectural discussions across the team as it grew from 4 to 10 engineers.

Senior Software Architect

Paramo Technology — Oct 2020 – Sep 2021 — Remote

- Owned the end-to-end architecture of a high-traffic online casino (iGaming) platform handling real-time money operations and regulatory reporting, working directly with the CTO.
- Spotted the issues slowing teams down and designed action plans that raised productivity by 40%, measured by features shipped and fewer errors.
- Designed and built event-driven microservices in .NET (C#, ASP.NET Core) for real-time game state, wallet operations, and regulatory reporting with strict consistency requirements; migrated relational databases to MongoDB where it made sense.
- Implemented distributed tracing across all services (OpenTelemetry, Application Insights, Datadog) to catch bugs early and resolve production issues faster.
- Brought in Kubernetes to scale; built automated deploy pipelines; helped migrate services to newer .NET versions.

- Applied object-oriented design, Domain-Driven Design (DDD) — aggregates, value objects, domain events — design patterns, and CQRS for read / write separation.
- Led part of the hiring and designed the technical entry exam; set engineering standards and delivered internal training on architecture patterns, design patterns, testing discipline, and operational best practices.

Software Architect

CloudX (Software Consultancy) — Sep 2017 – Oct 2020 — Remote / Buenos Aires (first employee)

- Was the company's first employee; led architecture decisions across several enterprise product lines, often working on-site with clients, and helped build part of the team as it grew.
- Built microservices backends in Node.js / TypeScript and .NET (C#, ASP.NET) across multiple enterprise product lines using Domain-Driven Design and Clean Architecture / Hexagonal patterns.
- Built a distributed system over geospatial data with PostGIS for a UK client.
- Built reusable Angular and React frontends with shared component libraries and TypeScript across enterprise products.
- Delivered technical training and conference talks to partners and internal engineering teams; collaborated with clients on requirements gathering and technical adoption.

Software Engineer (Angular / .NET)

Majestic Hotel Group — Jan 2014 – Aug 2017 — Argentina

- Built full-stack features in Angular and .NET for internal hotel management platforms, first as Team Lead and later as architect; led the migration from AngularJS to Angular 2.
- Redesigned the data model toward an event-oriented architecture, reducing coupling across integrations.

.NET Web Developer

Plus54 — 2011 – 2014 — Buenos Aires (earlier: Acsys 2009–2011, IBM 2008–2009)

- Early career: built .NET web applications, internal tooling, and mobile components end-to-end; integrated external APIs with unit testing and reliable data exchange patterns.

KEY ACHIEVEMENTS

Microsoft MVP, Developer Technologies — 8 consecutive years (2017–present), recognized for technical content and community contributions across the .NET and Microsoft ecosystem.

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) — Senior-level Azure architecture credential covering identity, governance, monitoring, data, business continuity, and infrastructure design.

Net-Baires Founder — Founded and grew Argentina's largest .NET developer community to 100K+ members across 10+ years.

80+ International Technical Talks — Conferences and meetups across LATAM and Europe on backend systems, architecture, AI, and software design.

Production Distributed Systems at Borderless — Built Rust and Python pipelines processing $50M+ in daily on-chain flow and contributing $100K+ in attributable annual P&L with execution under 10ms.

Polkadot Builder Program Graduate — Completed Polkadot Blockchain Academy in Lisbon, Portugal (2026).

Microsoft Student Partner — 5 years (2016–2021).

EDUCATION & CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) — Microsoft

Microsoft Certified: Azure Associate (AZ-104 / AZ-204 level) — Microsoft

Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)

Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)

Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)

OPEN SOURCE & COMMUNITY

Net-Baires (.NET Community) — Founder (2015–present). Argentina's largest .NET developer community, 100K+ members, 10+ years of monthly meetups and conferences.

Patrones Para Mortales — Founder (2020–present). Educational content brand on software design patterns and architecture.

Crypto Para Mortales — Founder (2023–present). Spanish-language blockchain and crypto education conference.

YouTube channel (youtube.com/@germankuber) — technical deep-dives on .NET, AI, software architecture, Rust, and backend systems.

SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams